Attention Multiscale Network for Semantic Segmentation of Multimodal Remote Sensing Images

Zhen Ye, Yuan Li, Zhen Li, Huan Liu, Yuxiang Zhang, and Wei Li, *Senior Member, IEEE*

***Abstract*— Due to recent advancements in deep learning, techniques for urban structure extraction and semantic segmentation of multimodal remote sensing images have significant improvements. However, the challenge arises from the variable color intensity and complex texture of urban structures in optical images, particularly in buildings and roads. Fortunately, the light detection and ranging (LiDAR) images promote the task of developing an optimal multimodal fusion network that effectively leverages information from different modalities. In this article, we propose an attention multiscale network (AMSNet) for binary semantic segmentation tasks focused on building extraction, as well as multiclass semantic segmentation tasks, by integrating optical and LiDAR remote sensing images. AMSNet introduces two feature fusion modules—spatial scale adaptive fusion (S^2AF) and semantic guided fusion (SGF). S^2AF facilitates feature fusion between optical and LiDAR images within the same layer. This module contains a spatial scale selection strategy and an adaptive weight learning strategy, which enables the network to adaptively extract and intentionally select multiscale features from multimodal data. SGF addresses the semantic gap between different layered block features through semantic feature guidance strategy while achieving feature fusion. Furthermore, we introduce robust feature learning (RFL) to ensure the network robustness in rotation and variation in objects, making it resilient to images captured from different viewpoints and sensors. RFL incorporates point-to-point similarity learning strategy and multiscale feature reuse strategy. Experimental results on publicly available datasets demonstrate that AMSNet outperforms other state-of-the-art models. Extensive ablation studies further confirm the significance of all key components in the proposed approach. The source code of this method is available at https://github.com/B-LG-J/AMSNet.git.**

***Index Terms*— Building extraction, multimodal fusion, multiscale attention learning, remote sensing images, semantic segmentation.**

Received 5 February 2024; revised 18 July 2024 and 29 December 2024; accepted 4 February 2025. Date of current version 21 February 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62301044, in part by China Postdoctoral Science Foundation under Grant 2023M730265, in part by Beijing Natural Science Foundation under Grant JQ20021, and in part by the Insight Action Project under Grant GYH2023033. *(Corresponding author: Huan Liu.)*

Zhen Ye, Yuan Li, and Zhen Li are with the School of Electronics and Control Engineering, Chang'an University, Xi'an 710064, China (e-mail: yezhen525@126.com; ly612592@163.com; 2021232078@chd.edu.cn).

Huan Liu, Yuxiang Zhang, and Wei Li are with the School of Information and Electronics, Beijing Key Laboratory of Fractional Signals and Systems, Beijing Institute of Technology, Beijing 100081, China (e-mail: huanliu233@gmail.com; zyx829625@163.com; liwei089@ieee.org).

Digital Object Identifier 10.1109/TGRS.2025.3540848

I. Introduction

BUILDING extraction and semantic segmentation of remote sensing images play crucial roles in various applications, including urban planning, demographics, economic assessment, and disaster management [1], [2], [3], [4], [5]. The predominant image modality used for data capture in these tasks is optical imagery [6], [7]. However, due to the diverse structural characteristics of urban landscapes and the significant variability among different buildings and roads, relying solely on optical images poses challenges in achieving clear recognition of urban features. Consequently, researchers have shifted their focus to joint use of multiple modal remote sensing images. Light detection and ranging (LiDAR) images, captured by an active sensor, provide digital surface model (DSM) data that incorporate elevation information [8]. Leveraging the strengths of DSM and the high-resolution details offered by optical images, these two modalities complement each other in extracting ground objects, particularly buildings. Optical images reveal intricate details of urban areas, while LiDAR supplies crucial height information. Therefore, researchers advocate for integrating these two modalities to extract complementary surface data [9], [10], [11].

To effectively integrate two modalities, the primary consideration in architecture design should be how to adequately fuse the complementary features of multimodal data. Some methods use splicing-based techniques [12], [13], [14], while others use summing-based approaches [15] or attentional mechanisms [16]. However, these methods assign equal importance to all multimodal data without considering that complementary data may vary in importance across different attributes. To address this issue, some methods use gating mechanisms [17], [18], which can adaptively assign different weights to different attributes of the multimodal data during the feature fusion process. Nevertheless, these methods often perform multimodal feature fusion at a single scale, neglecting the fact that multimodal data behave differently at different scales due to sensor resolution differences. This can result in incomplete extraction of the complementary properties of the multimodal data. In addition, these methods typically use linear upsampling techniques to inject contextual information from high-level feature maps into low-level ones, which may cause feature discontinuity and object edge blurring. Therefore, to fully use contextual information and bridge the

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semantic gaps between different feature levels, improvements in upsampling methods are necessary.

To summarize, although the aforementioned multimodal methods for building extraction and semantic segmentation demonstrate better performance compared with single-modal methods, several issues remain.

1) Due to sensor differences in spatial resolution, multimodal data exhibit different levels of complementary information at different scales (e.g., the amount of information from optical and LiDAR remote sensing data at the same scale can vary). Therefore, multimodal data fusion needs to consider features at multiple spatial scales, a factor overlooked by most existing multimodal methods.
2) Exploiting high-level contextual information is crucial for semantic segmentation tasks. Spatial dimension alignment is required to obtain this information. However, most existing methods use upsampling for spatial dimension alignment, which may produce intermittence and blurriness, making it difficult to capture rich semantic information.
3) In multimodal feature extraction (FE), since the data come from different viewpoints and sensors, the target objects may have different sizes and poses. Therefore, the network needs better robustness to rotation and variation, a requirement not addressed by most existing multimodal FE strategies.

To address the issues mentioned above, we present a novel attention multiscale network (AMSNet). This network is designed to selectively emphasize crucial information from optical and LiDAR remote sensing data across various scales for adaptive feature fusion. First, the spatial scale adaptive fusion (S^2AF) module is introduced to capture important information from multimodal data at multiple scales through a combination of spatial scale selection learning strategy and adaptive weight learning strategy. To alleviate the problem of feature discontinuity and object edge blurring, we propose the semantic guided fusion (SGF) module, which decodes multimodal features layer by layer using an upper level and lower level feature cross-alignment learning strategy, along with an attention learning strategy. To enhance the robustness of multimodal data against rotation and variation in objects, we use a point-to-point similarity learning strategy and a multimodal feature reuse strategy. These strategies help mitigate the issue of target objects manifesting in various sizes and poses. Overall, this approach ultimately enhances the accuracy of building extraction and semantic segmentation using optical and LiDAR data.

The main contributions can be summarized as follows.

1) A novel AMSNet for building extraction and semantic segmentation (AMSNet) is introduced in this article. To harness the inherent characteristics of multimodal data effectively, we use a dual-branch structure during the encoding stage, integrating optical and LiDAR data. This strategic design facilitates the exhaustive exploration of complementary features inherent in both the optical and LiDAR modalities. Subsequently, during the decoding stage, fusion information is intricately modeled with contextual information, enabling the network to glean global insights and information.
2) S^2AF and SGF modules are introduced to facilitate the integration of intralayer multimodal features and to unify interlayer deep and shallow features, respectively. In the context of interlayer information fusion, an adaptive spatial selection strategy is used to select the most important features in each spatial scale and an adaptive learning strategy is used to assign weights to the multimodal data based on the importance of their attributes. Conversely, for information fusion between distinct layers, the approach of interlayer semantic information guidance strategy is considered. This strategy ensures the network progresses layer by layer with precision, mitigating the problem of intermittent phenomena and blurring of object edges, incorporating and refining semantic information at each step.
3) The proposed robust feature learning (RFL) module aims to enhance the network's resilience against rotation and variation. By doing so, it significantly improves the network's proficiency in recognizing objects with various scales and poses. This augmentation in robustness ensures that the network maintains its accuracy and effectiveness, leading to more reliable and versatile object recognition capabilities.

The remainder of this article is organized as follows. Section II describes related work. Section III describes the proposed AMSNet. Section IV describes experimental data and setup. Section V discusses the results and analysis. We conclude this article in Section VI.

II. Related Work

A. Attention Mechanism

Attention mechanism has become a widely adopted technique for enhancing the accuracy of image semantic segmentation. It autonomously identifies crucial information, thereby amplifying semantic expression. Typically, two major approaches based on attention mechanism are observed, including the gradient-descent-based approach and the reinforcement-learning-based method. The gradient-descent-based approach relies on the objective function and the corresponding optimization function. Chen et al. [19] integrate spatial and channel attention (SCA) seamlessly into the convolutional neural network (CNN). The proposed SCA-CNN not only incorporates spatial and channel attention but also uses a multilayer attention mechanism. Fu et al. [20] introduce a dual-relation-aware attention network (DRANet) using positional attention and channel attention, yielding favorable outcomes when applied to semantic segmentation. Motivated by a Reward function [21], the reinforcement-learning-based methods exhibit attention mechanism by compelling the model to concentrate on specific local details. The recurrent models of visual co-attention (RMCA) model proposed by Mnih et al. [22] use an attention mechanism based on a reinforcement learning approach, compelling the model to concentrate on specific local details and training the model with a Reward function. It is characterized by the feature of not focusing on the entire image at all times but rather

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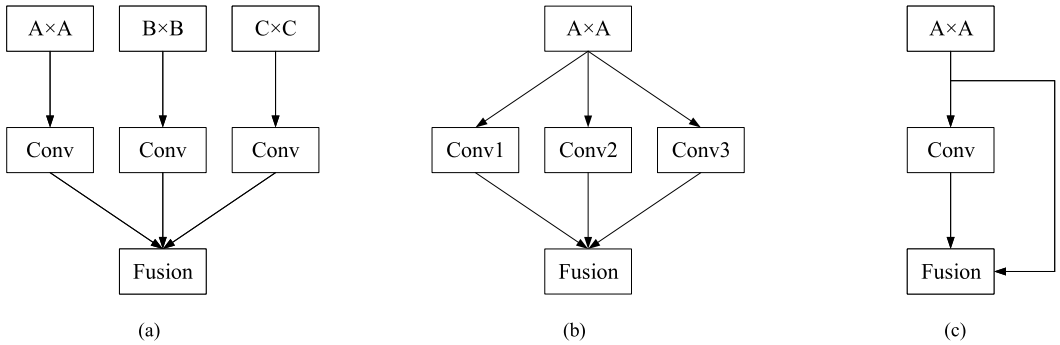


Fig. 1. Three multiscale learning methods. (a) Inputs of different sizes. (b) Convolutional kernels of different sizes. (c) Integration of features from different locations.

sweeping over the image in a specific sequence. The recurrent attention CNN (RACNN) proposed by Fu et al. [23] recursively analyzes local information, extracting essential features. The incorporation of the attention mechanism ensures that the entire network not only concentrates on overall information but also pays attention to local details.

B. Multiscale Learning

Various surfaces in remote sensing images encompass feature types of diverse sizes and shapes, necessitating a pivotal role for multiscale learning in feature representation. Presently, the existing multiscale learning methods can be broadly categorized into three groups: 1) incorporating images of different sizes as input; 2) constructing multibranch structures with convolution kernels of varying sizes on each branch; and 3) merging features from different levels. The schematic representation of these three multiscale FE methods is illustrated in Fig. 1. In the GoogLeNet network, Szegedy et al. [24] introduced the Inception module, using a parallel multibranching strategy. It uses convolutional kernels of different sizes to extract extra-multiscale features from the same input feature map. Subsequent evolutionary versions, including Inception V1-V3 [24], [25] and Inception-V4 [26], were developed based on this concept. He et al. [27] proposed a spatial pyramid pooling structure, incorporating multiple pooling layers to extract input features at various scales. Furthermore, the introduction of Res2Net has enhanced multiscale feature representation through a hierarchical residual structure, allowing for a more refined representation of multiscale features. This improvement significantly expands the receptive fields of the network [28].

C. Multimodal Information Fusion

Multimodel information fusion involves combining information from multiple sources, which aims to achieve a coherent explanation or description of the object under investigation, enhancing the overall performance of the information system. Fusion models typically operate at three levels: data, feature, and decision. Audebert et al. [29] delved into the foundations of FuseNet and extensively explored two multimodal fusion methods falling under feature fusion. The study's outcomes reveal that both the fusion structures effectively leverage the advantages of multimodal data. In a different approach, Paisitkriangkrai et al. [30] combined CNN with handcrafted designed features to construct a semantic segmentation model using both red–green–blue (RGB) images and DSM images. This method aligns with data fusion. Subsequently, Sherrah [31] proposed a semantic segmentation fusion network based on optical images and DSM images, representing a decision fusion approach.

III. Methodology

In this section, we present the proposed AMSNet with an overview outlined in Fig. 2. Initially, optical and LiDAR remote sensing images are input into two distinct network streams, using ResNet-34 as the backbone encoder. The encoder architecture is enhanced with an S^2AF module and an RFL module. The S^2AF module is designed to dynamically select the best multimodal information with multiple scales. Simultaneously, the RFL module is incorporated to cultivate rotation robustness and variation robustness. This enhancement equips the network to effectively address challenges posed by the diverse sizes and poses of target objects in multimodal remote sensing data, thereby elevating the network's multimodal information extraction capabilities. The decoder stream uses the SGF module to systematically decode multimodal information layer by layer. This involves establishing contextual dependencies guided by semantic information, facilitating the learning of global features, and ultimately generating the final prediction map.

A. Network Architecture

As illustrated in Fig. 2, both the optical and the LiDAR FE branches undergo an identical encoding process except that the input of the LiDAR branch is single-channel input while that of the optical branch is three-channel. The initial encoder block encompasses a 7×7 convolutional layer with a stride of 2, followed by a batch normalization (BN) layer and a rectified linear unit (ReLU) layer. Subsequently, the second encoder block includes a 3×3 maximum pooling layer with a stride of 2 and incorporates the initial residual layer of ResNet-34. The subsequent three residual layers are sequentially added. Except for the first residual layer, all three incorporate a residual unit that downsamples the feature map and doubles the feature channels. Consequently, the resolution of the final output produced by the encoding process is 1/32 of that of the original input.

In the decoder phase, the fusion information is decoded, and the SGF module is used to ensure that the size of the final output aligns with the size of the original input. Alongside the

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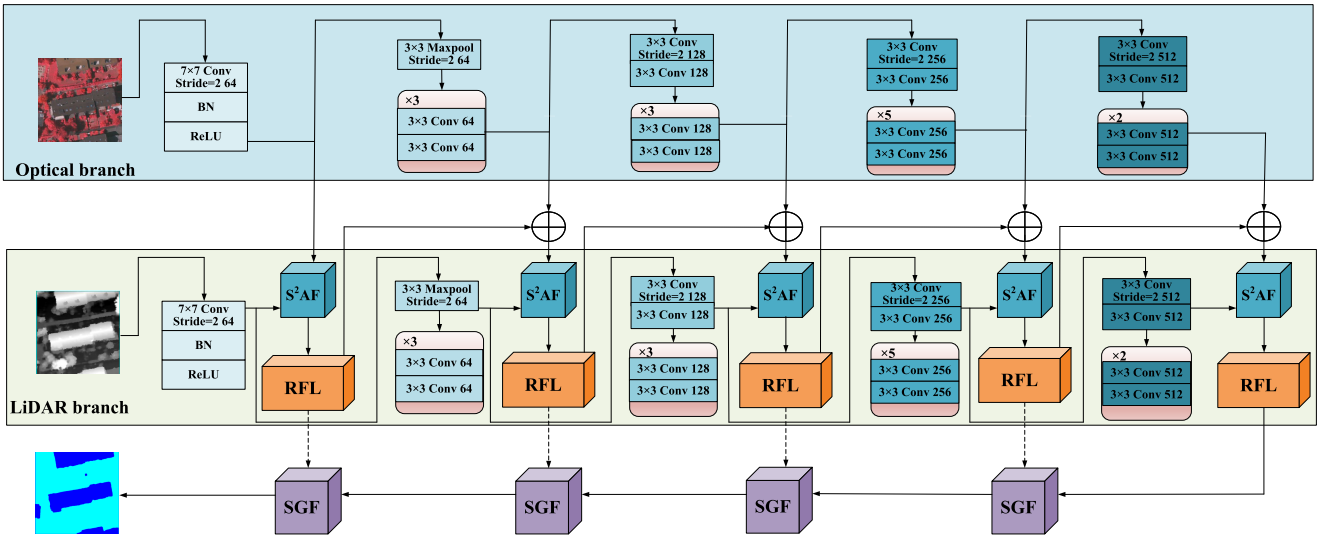


Fig. 2. Overall framework of the proposed AMSNet, which is based on an encoder–decoder architecture and contains three main modules to adequately extract complementary features of optical and LiDAR data for semantic segmentation tasks. The main modules include S^{2}AF, RFL, and SGF.

Algorithm 1 Pseudocode for Our Method Training

Input: optical data X_h, LiDAR data X_l, Ground truth Y, number of training epochs.
Output: Classification map O.
Parameter setting and weights initialization.
for $epoch < epochs$ **do**
Extract multimodal fusion features $L^{(m)}$ of optical data and LiDAR data according to (1), where m represents layer order of encoder.
Feed $L^{(m)}$ into RFL module to obtain $D^{(m)}$. Next, Enter $L^{(m)}$, $D^{(m)}$ into the SGF module. Then, classification map will be obtained by (3).
Update parameters and weights via Adamax optimizer.
end for
Obtain the probability distribution and classification map.

backbone encoder–decoder structure outlined previously, this section incorporates three specifically designed modules.

The majority of the existing multimodal fusion methods for optical and LiDAR rely on techniques such as splicing, summing, or adaptive methods [32]. However, these approaches tend to overlook the problem that multimodal data present different degrees of complementary information with multiple scales due to differences in sensor spatial resolution. To provide comprehensive learning of complementary features, an innovative S^{2}AF module is introduced in this article. Let $L^{(m)}$ represent the output feature of the mth layer of the S^{2}AF, expressed as

$$L^m = \mathrm{S}^2\mathrm{AF}\left(E^{(m)}_{\mathrm{optical}}, E^{(m)}_{\mathrm{dsm}}\right), \quad m = 1, \ldots, 5 \tag{1}$$

where S^{2}AF represents the spatial scale adaptive fusion module, $E^{(m)}_{\mathrm{optical}}$ signifies the output feature of the mth encoder in the optical branch, and $E^{(m)}_{\mathrm{dsm}}$ indicates the output feature of the mth encoder in the LiDAR branch.

In multimodal FE, the target object may exhibit various sizes and poses due to diverse data sources, necessitating a high level of robustness in the network. Convolutional kernels with small sizes are sensitive to the spatial rotation of the input, impacting the network's rotational invariance and FE accuracy [17]. Conversely, larger convolutional kernels can enhance the network's rotational invariance, but they incur a high computational cost, rendering the network less efficient. To address this challenge, this article introduces the RFL module. Let $D^{(m)}$ represent the output of the mth layer, expressed as

$$D^m = \mathrm{RFL}(L^m), \quad m = 1, \ldots, 5 \tag{2}$$

where RFL denotes the RFL module.

Given the importance of global contextual information in building extraction and semantic segmentation of multimodal remote sensing data, this article incorporates the SGF module. The SGF module is used in the decoder stage to facilitate the learning of global information from multimodal data and model contextual dependencies at various layers. By integrating the SGF module into the decoder network, it enables the extraction of feature mapping for global information. The output O of the decoder branch can be represented as

$$O^n = \mathrm{SGF}\left(D^m, D^{m-1}\right), \quad n = 5 - m \tag{3}$$

where SGF denotes the semantically guided fusion module and $O^{(n)}$ denotes the output of the nth layer decoder.

B. S^{2}AF Module

Multimodal information exhibits varying performance capabilities at different spatial scales, and extracting the most crucial multimodal features at each scale can significantly enhance the effectiveness of multimodal fusion. The existing approaches for joint use of optical and LiDAR data often neglect this aspect. To address this, this article introduces the S^{2}AF module, using a multiscale interactive approach

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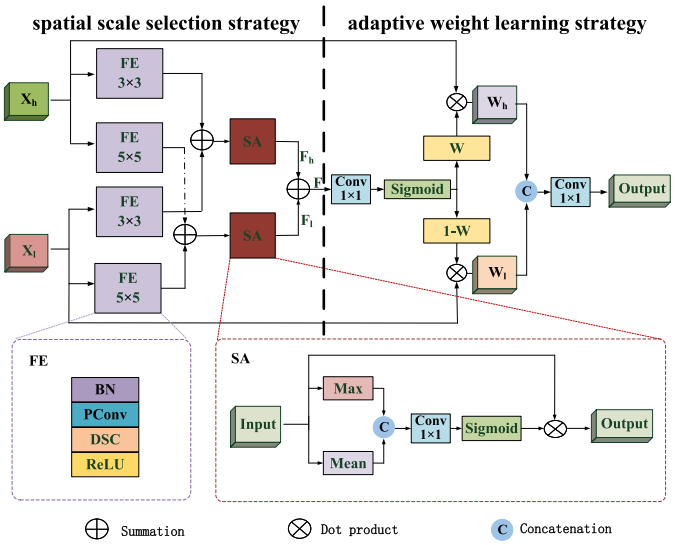


Fig. 3. S^{2}AF module, including spatial scale selection strategy and adaptive weight learning strategy.

to extract important multimodal features at different spatial scales, particularly beneficial for accurately capturing buildings covered by vegetation, as illustrated in Fig. 3.

The process begins with FE blocks at different scales (convolutions with kernel sizes 3×3 and 5×5 for both the optical and LiDAR branches). These blocks include BN, partial convolution (PConv), depthwise separable convolution (DSC), and ReLU in a sequential fashion. Following FE, the features of different modalities (LiDAR-branch features X_l and optical-branch features X_h) at the same scale undergo further processing using the spatial attention (SA) block to select important multimodal information. Next, a 1×1 convolution is applied to calculate the correlation of the features. The entire process can be expressed as

$$\begin{aligned} F_h &= \mathrm{SA}(\mathrm{FE}_{3\times3}(X_h) + \mathrm{FE}_{3\times3}(X_l)) \\ F_l &= \mathrm{SA}(\mathrm{FE}_{5\times5}(X_h) + FE_{5\times5}(X_l)) \\ F &= F_h + F_l \\ \mathrm{FE}_{3\times3} &= \mathrm{BN}(PConv(D_{3\times3}(\mathrm{ReLU}(B)))) \\ \mathrm{FE}_{5\times5} &= \mathrm{BN}(PConv(D_{5\times5}(\mathrm{ReLU}(B)))) \\ \mathrm{SA} &= S\big(f_{1\times1}(C[\mathrm{Mean}(A), \mathrm{MAX}(A)])\big) \end{aligned} \tag{4}$$

where A represents F_h or F_l and B represents X_h or X_l; $D_{3\times3}$ and $D_{5\times5}$ denote 3×3 and 5×5 depthwise separable convolutions, respectively.

Subsequently, the Sigmoid activation is used to obtain the weighted probability matrix W, and the features X_l and X_h are multiplied by $(1 - W)$ and W, respectively. This results in the corresponding weighted features W_l and W_h. Finally, these features are concatenated using 1×1 convolution to ensure consistency in input and output dimensions. The entire process can be expressed as

$$\begin{aligned} W &= S(f_{1\times1}(F)) \\ F_{\text{fusion}} &= f_{1\times1}(C[WX_h, (1-W)X_l]) \end{aligned} \tag{5}$$

where S stands for Sigmoid, $C[\cdot]$ stands for concatenation, and $f_{1\times1}$ stands for 1×1 convolution.

The main innovations of this module can be summarized in the following two points. First, an adaptive spatial scale selection strategy is used to identify crucial features for optical and LiDAR data. This adaptive spatial scale selection uses FE blocks to extract multimodal features at varying scales and SA blocks assign adaptive importance weights to these features, facilitating the optimal selection of them. Second, adaptive learning strategy is incorporated to address the challenge of multimodal information having diverse impacts on urban structures at different locations (e.g., LiDAR data are more important for elevated buildings, while optical data are more effective for objects on the ground, such as roads). The proposed network dynamically and adaptively assigns weights to enhance the performance of building extraction and semantic segmentation.

C. RFL Module

Given that the multimodal data originate from distinct viewpoints and sensors, and the target objects may vary in sizes and poses, the network is required to exhibit robustness in rotation and variable scales of objects. A small-size convolutional kernel, while robust to small object sizes, is sensitive to the spatial rotation of inputs, impacting the rotational robustness of network and its ability to handle large-size objects. On the other hand, large-size convolutional kernels can enhance rotational robustness and handle large-size objects but at the cost of increased computational complexity, resulting in an inefficient network. To address these challenges, this article proposes a multiscale strategy to ensure robustness in rotation and variable scales of objects, thereby improving the accuracy of the multimodal network. To preserve the spatial features of optical data, we use transposed convolution instead of regular convolution. This choice allows for the restoration of a certain amount of detail information before downsampling and compression, enhancing image quality and clarity. This approach aims to retain more spatial details while ensuring rotational robustness. The specific workflow of this module is depicted in Fig. 4 (the figure illustrates the RFL of the first layer of the encoder as an example, with 64 input channels). Initially, the input feature map undergoes a series of 1×1 convolutions and Hessian filters. The 1×1 convolution reduces the number of feature map channels (changing from 64 to 32), decreasing model complexity while retaining essential feature information. Simultaneously, Hessian filters process the downscaled feature map to further extract key texture and shape information. Following this, each RFL layer divides the input feature map equally along the channel dimension into four partitions (the channel dimension of each partition is 8), i.e.,

$$Z_1, Z_2, Z_3, Z_4 = \mathrm{Split}(Z). \tag{6}$$

Subsequently, the first input partition Z_1 preserves the original features without any processing. The second input partition Z_2 undergoes transpose convolution and a point-to-point similarity calculation. This allows the network to extract spatial features while leveraging pixel similarity to aggregate features with high similarity, thereby reducing the network sensitivity to rotational information. This process yields the output feature

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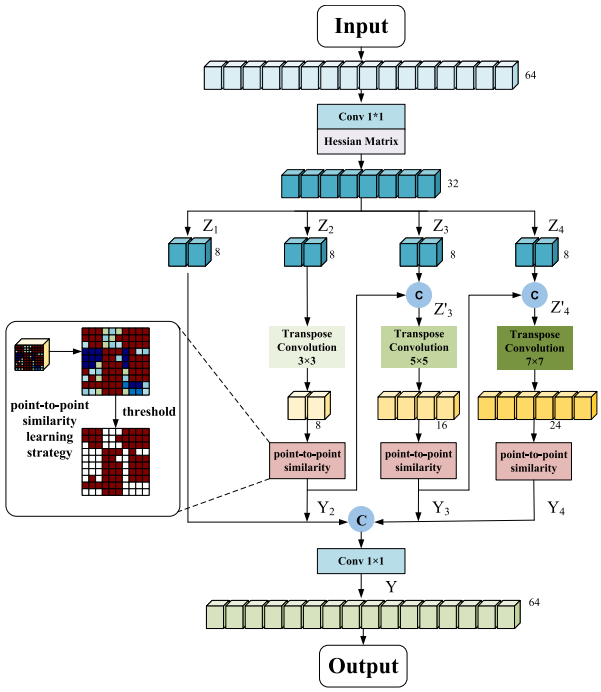


Fig. 4. RFL module, including point-to-point similarity learning strategy and multiscale feature reuse strategy.

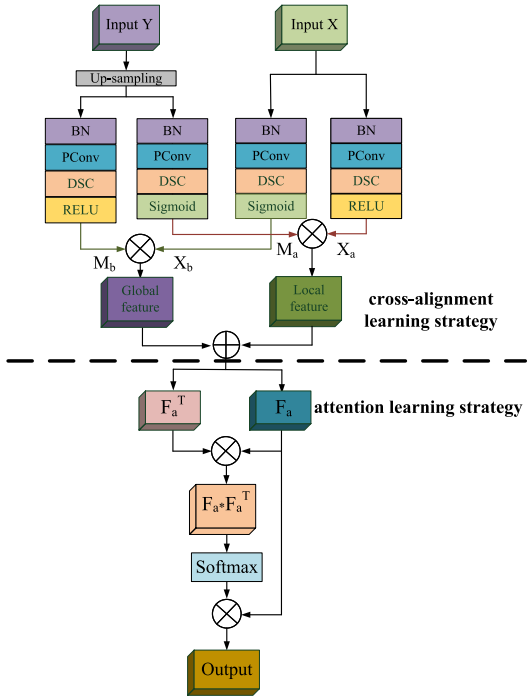


Fig. 5. SGF module, including cross-alignment learning strategy and attention learning strategy.

map Y_2

$$F = \text{Transpose}(Z_2)$$
$$Y_2 = P(F) \odot F \tag{7}$$

where Transpose represents the transpose convolution, and the element $P(F)_{ij}$ of $P(F)$ is the number of pixels in F that are similar to the pixel F_{ij} and each pixel F_{mn} satisfies $\exp(\sum_k (F_{ij,k} - F_{mn,k})^2) > T$, where T is a threshold that is fixed as 0.6 in the experiment.

Next, we concatenate Y_2 with the third input partition Z_3, resulting in a channel dimension of 16. This channelwise concatenation effectively achieves feature reuse. The resulting concatenated feature map Z_3' is then processed through transpose convolution and a point-to-point similarity calculation to extract multiscale similarity features. This process is outlined as follows:

$$Y_3 = P\left(T\left(Z_3'\right)\right) \odot T\left(Z_3'\right). \tag{8}$$

Following that, we concatenate Y_3 with Z_4, producing cascaded features Z_4' with 24 channels. Z_4' undergoes transpose convolution and point-to-point similarity calculation once again to obtain Y_4, where

$$Y_4 = P\left(T\left(Z_4'\right)\right) \odot T\left(Z_4'\right). \tag{9}$$

Finally, we concatenate Z_1, Y_2, Y_3, and Y_4 and then generate cascade features Y, where

$$Y = f_{1\times1}(C[Z_1, Y_2, Y_3, Y_4]). \tag{10}$$

The key innovations of this module can be outlined as follows. First, the adoption of a multiscale technique ensures initial robustness in handling both rotation and variation. Second, the integration of a Hessian filter serves to detect features with rotational invariance by calculating gradient and curvature information. This inclusion aims to introduce rotational invariance into the network. Finally, the exploration of pixel similarity using Euclidean distance ensures that the spatial features extracted by the RFL module are independent of the spatial rotation of multimodal features. This independence arises from the fact that pixel similarity is not contingent on the spatial location of the pixels.

D. Semantic Guidance Fusion Module

Learning semantic information from optical and LiDAR remote sensing images proves advantageous for enhancing the performance of semantic segmentation networks. In addition, considering the contextual relationship of the target object is crucial for obtaining comprehensive and accurate segmentation results. In light of these considerations, we propose the SGF module. The SGF module uses upper level and lower level feature cross-alignment learning strategy, along with attention learning strategy, to extract rich contextual information from hierarchical features. This leads to the establishment of a network structure that prioritizes accurate semantic information, particularly tailored for the global extraction of semantic details.

Fig. 5 demonstrates the relative nature of local and global features. For example, in Fig. 2, the output from the fourth RFL layer (input X) mainly captures local details, while the output from the fifth RFL layer (input Y) encompasses broader global characteristics. As the network deepens, semantic information progressively intensifies, while spatial information is relatively diluted. In addition, the input for the SGF module is sourced from the outputs of two adjacent layers of the decoder. Upsampling of the lower layer's output ensures dimensional alignment with the output from the upper layer. Subsequently, basic features of the adjacent layers are extracted using a

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series of operations including BN, PConv [33], DSC, and ReLU. Simultaneously, the weights of the adjacent layers are extracted through a series of operations involving BN, PConv, DSC, and Sigmoid. To facilitate interactive learning, dot multiplication is applied to the basic features of adjacent layers with their corresponding weights. This process generates local features for the upper layer and global features for the lower layer. The two sets of features are then summed to achieve interactive learning of semantic information between the upper and lower layers of the decoder, effectively reducing the semantic gaps between features in cross-layer blocks. Finally, the aligned feature map (F_a) and its transposition (F_a^T) are fed into the Softmax to perform a weighted summation of F_a, thereby achieving semantic information-driven selection of important semantic features. This process can be represented as

$$\begin{aligned} X_{\text{local}} &= \sigma(D_{3\times3}(P(\text{BN}(X)))) \\ W_{\text{local}} &= S(D_{3\times3}(P(\text{BN}(X)))) \\ X_{\text{global}} &= \sigma(D_{3\times3}(P(\text{BN}(\text{UP}(Y))))) \\ W_{\text{global}} &= S(D_{3\times3}(P(\text{BN}(\text{UP}(Y))))) \\ F_\alpha &= X_{\text{local}} \times W_{\text{global}} + W_{\text{global}} \times X_{\text{local}} \\ \text{Output}_{\text{SGF}} &= \text{softmax}\left(F_\alpha F_\alpha^T\right) F_\alpha \end{aligned} \tag{11}$$

where σ denotes ReLU function and S denotes Sigmoid function.

The innovation of this module lies in its ability to learn the alignment of cross-layer features. When performing size alignment, a technique is used that uses PConv's mask to separate the valid and invalid regions. This approach retains the information of object edges and mitigates the issue of losing edge information through filling strategies. Consequently, this effectively addresses intermittent phenomena and blurring problems associated with object edges caused by the upsampling method. The module excels in learning rich semantic information by reducing the semantic gap between features of different decoding layers through cross-learning between upper and lower layers. Ultimately, attention is applied to learn the most important semantic information, facilitating layer-by-layer decoding of features dominated by precise semantic information.

IV. Experimental Data and Setup

A. Datasets

1) Potsdam[1]*:* The Potsdam dataset comprises 38 pairs of optical and LiDAR images. Each image has a spatial size of 6000 × 6000 with a spatial resolution of 0.05 m. After cropping, the dataset encompasses six primary semantic classes: *impervious surfaces, buildings, low vegetation, trees, car, and clutter*.

2) Vaihingen[2]*:* The Vaihingen dataset is composed of 33 pairs of optical and LiDAR images. Each image has a spatial size of 2500 × 2000, and the spatial resolution is 0.05 m. Similar to the Potsdam dataset, the Vaihingen dataset includes the same identified classes: *impervious surfaces, buildings, low vegetation, trees, car, and clutter*.

3) DFC2018[3]*:* The DFC2018 dataset comprises 14 pairs of optical and LiDAR images. Each image has a spatial size of 1192 × 1202 with a spatial resolution of 0.05 m. The original dataset encompasses 21 classes, namely, *unclassified buildings, residential buildings, nonresidential buildings, stadium seating, roads, pavements, pedestrian crossings, main arterials, highways, railways, trains, water, evergreens, deciduous trees, healthy grass, stressed grass, artificial turf, paved car parks, unscored car parks, cars, and bare soil*.

B. Experimental Setup

1) Implementation Details: All the experiments are carried out on a single NVIDIA GeForce RTX 3090 GPU using the PyTorch framework. Due to GPU memory limitations, images are cropped using a 640 × 640 sliding window with no overlap. After cropping and selecting images containing buildings, the numbers of image pairs in the Potsdam, Vaihingen, and DFC2018 datasets were 398, 1415, and 14, respectively. For the Potsdam dataset, 50% of the image pairs were designated as the training set, 20% as the validation set, and 30% as the test set. In the case of the Vaihingen dataset, 80% of the image pairs were allocated to the training set, 10% to the validation set, and 10% to the test set. Similarly, for the DFC2018 dataset, 80% of the image pairs were assigned to the training set, 10% to the validation set, and 10% to the test set.

In all the experiments, the cross-entropy loss function is used to optimize the model during training. This choice ensures that the evaluation of the proposed method focuses solely on the contributions of the network architecture without introducing additional complexities from the loss function.

All the methods used in the subsequent experiments share the same batch size and number of epochs. To address overfitting and artificially boost the effective training sample size [17], data augmentation is applied. This involves random rotations of the input images in both the vertical and horizontal directions for all the methods. The AdaMax algorithm [34] is used for optimization with a weight decay set to 0.0009. Optimization of the learning rate is performed using a "poly" policy [35]. For all the networks, the initial learning rate is set to 0.001 and is multiplied by $(1 - (\text{iter}/\text{max-iter}))^{\text{power}}$ during training. Here, the "power" parameter is set to 0.3, and "max-iter" is calculated by multiplying the number of epochs with the total number of batches.

2) Evaluation Metrics: This article evaluates the networks using three widely used metrics: overall accuracy (OA), mean intersection over union (MIoU), and frequency weighted intersection over union (FWIoU).

1) *OA:* Proportion of correctly classified pixels or regions to all pixels or regions.
2) *MIoU:* This is a commonly used measure for semantic segmentation which involves measuring the overlap between the predicted labels and the ground-truth labels, and the final evaluation metric is obtained by averaging the IoUs across all the targets.

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[3]https://hyperspectral.ee.uh.edu/?page_id=1075

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TABLE I
EVALUATION OF METRIC SCORES (%) OF BUILDING EXTRACTION FOR THREE DATASETS

Dataset	Metrics	Single-branch networks				Dual-branch networks					
		ResUNet++ [36]	PSPNet [37]	Deeplab V3+ [38]	AMSNet-single	FuseNet [39]	RedNet [40]	HAFNet [16]	$(MS)^2$-Net [41]	MCANet [42]	AMSNet
Potsdam	OA	95.6±1.927	96.8±0.557	97.5±0.469	97.6±0.334	95.8±1.592	96.3±0.271	97.0±0.350	97.6±0.124	96.0±0.498	**99.1±0.104**
	FWIOU	91.7±3.315	94.3±0.682	95.2±0.894	95.5±1.093	91.9±3.030	93.7±0.354	94.5±0.652	95.5±0.235	92.4±0.564	**97.9±0.276**
	MIOU	87.2±7.330	91.4±1.072	94.1±1.524	94.8±1.005	87.6±4.824	90.1±1.130	91.5±1.072	94.9±0.264	91.7±1.471	**97.4±0.377**
Vaihingen	OA	94.1±1.053	94.0±1.261	96.2±1.398	96.8±0.092	96.1±0.251	96.8±0.221	96.4±0.145	95.3±0.990	96.5±0.764	**98.5±0.137**
	FWIOU	90.7±1.988	89.6±2.292	93.0±2.386	93.9±0.112	92.5±0.445	94.5±0.125	93.5±0.272	92.1±1.810	93.3±0.439	**97.0±0.198**
	MIOU	89.0±2.448	88.8±4.445	92.0±4.465	92.7±0.204	91.1±0.502	92.4±0.502	92.1±0.323	90.8±1.906	92.5±1.136	**96.4±0.278**
DFC2018	OA	91.6±1.023	88.3±1.759	86.0±0.602	93.5±0.145	83.2±1.207	95.6±0.114	93.7±0.218	86.4±0.300	96.8±0.098	**98.9±0.238**
	FWIOU	85.4±1.985	82.1±2.095	79.6±1.167	87.3±0.374	77.8±1.749	92.7±0.275	89.7±0.370	80.2±0.595	93.9±0.239	**97.7±0.541**
	MIOU	75.0±2.324	73.1±2.823	65.6±2.015	78.9±0.660	55.3±1.940	85.3±0.720	81.5±0.467	66.7±1.598	89.5±0.579	**94.2±0.774**

3) *FWIoU:* This version of MIoU, known as FWIoU, introduces the concept of assigning weights to classes based on their occurrence frequencies. In this approach, the IoUs of different classes are multiplied by their respective weights and the resulting products are summed to obtain the FWIoU metric. By considering the frequency of each class, FWIoU provides a more balanced evaluation that takes into account the importance of each class in the overall segmentation task.

V. RESULTS AND ANALYSIS

A. Comparison to State-of-the-Art Methods

We compare the proposed AMSNet with several state-of-the-art two-branch deep semantic segmentation models, such as FuseNet [39], RedNet [40], HAFNet [16], $(MS)^2$-Net [41], and MCANet [42] to provide a comprehensive performance evaluation of AMSNet. In addition, due to the small amount of open source code for two-branch networks, and to reflect the reliability of the experiments, we simplified AMSNet so that it is also compared with single-branch semantic segmentation models (i.e., ResUNet++ [36], PSPNet [37], and Deeplabv3+ [38]). The simplified AMSNet structure is described as follows, first, the encoder structure of AMSNet is simplified by eliminating the use of the S^2AF module, allowing the downsampled features extracted by the ResNet-34 encoder at the fifth layer to be transmitted directly to the RFL module. In the decoder stage, the outputs of the two neighboring RFL modules are fed into the SGF module as in a normal AMSNet network, resulting in a semantic segmentation network containing only optical branch. This simplified network is called AMSNet-single.

Table I presents the experimental results of various methods for building extraction, and it can be observed that the proposed AMSNet outperforms all other competing methods in terms of OA, FWIoU, and MIoU. For the Potsdam dataset, the margins of OA, FWIoU, and MIoU of AMSNet over the second-best model, $(MS)^2$-Net, are 1.5%, 2.4%, and 2.5%, respectively. For the Vaihingen dataset, the OA, FWIoU, and MIoU of AMSNet yield 1.7%, 2.5%, and 4.0% better performance compared with the best dual-branch competitor RedNet. For the DFC2018 dataset, the OA, FWIoU, and MIoU of AMSNet are notably higher than those of MCANet by 2.1%, 3.8%, and 4.7%, respectively. In addition, the metrics of AMSNet-single also have the highest scores among single-branch methods. AMSNet achieves significant performance improvements over AMSNet-single across all three datasets, demonstrating the critical role of LiDAR data in building extraction. For the Potsdam dataset, OA, FWIoU, and MIoU improve by 1.5%, 2.4%, and 2.6%, respectively. For the Vaihingen dataset, the improvements are 1.7%, 3.1%, and 3.7%, respectively. For the more complex DFC2018 dataset, the improvements are the most significant, reaching 5.4%, 10.4%, and 15.3%, respectively.

The results of all the models for building extraction on all three datasets are shown in Figs. 6–8. From these three sets of figures, it can be seen that all the models can obtain good extraction results when performing building extraction on large areas. However, the robustness and effectiveness of all the compared methods are not good enough for some small buildings or the edge cases of buildings. The red rectangular box in the first row of Fig. 6 for the Potsdam dataset shows that only the proposed AMSNet closely agrees with the contour information of the real scenario on the ground. Similarly, in the second row of Fig. 7 for the Vaihingen dataset, the red rectangular box indicates that AMSNet also aligns well with the contour information. In addition, in the DFC2018 dataset, shown as Fig. 8, the red box on the right side of the third row demonstrates that AMSNet exhibits the closest agreement with the contour information for the extraction of tiny buildings.

To further assess the performance of the proposed AMSNet, precision–recall (P–R) experiments are conducted. Precision measures the number of true positive predictions divided by the total number of positive class predictions, indicating how accurate the model's positive predictions are. On the other hand, recall measures the number of true positive predictions divided by the total number of actual positive instances, indicating how well the model captures all positive instances. So, precision reflects the accuracy of positive predictions from a prediction outcome perspective, while recall reflects the ability to recall positive class instances from a labeling perspective. Generally, if one model's P–R curve is enclosed by another model's curve or the area covered by the curve is larger, it indicates superior performance. As shown in Fig. 9, the proposed AMSNet exhibits better performance than other models for the three datasets.

To generalize the network structure to reflect the network structure, we also perform the semantic segmentation task. Table II shows the segmentation performance of the proposed AMSNet and other competing methods. It can be observed that the proposed AMSNet achieves the best OA, FWIoU,

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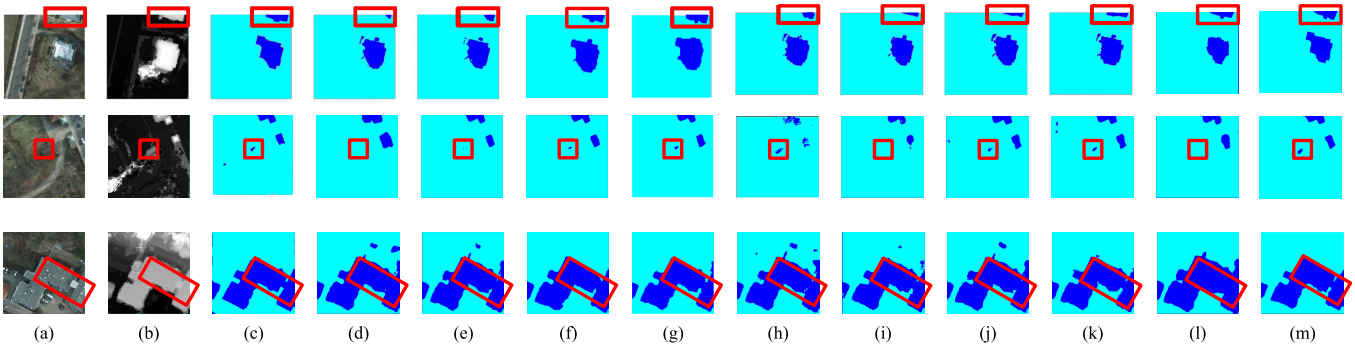


Fig. 6. Qualitative comparison of building extraction for different methods on the Potsdam dataset. (a) Optical. (b) LiDAR. (c) Ground truth. (d) ResUNet++ (95.6%). (e) PSPNet (96.8%). (f) Deeplab V3+ (97.5%). (g) AMSNet-single (97.6%). (h) FuseNet (95.8%). (i) RedNet (96.3%). (j) HAFNet (97.0%). (k) $(MS)^2$-Net (97.6%). (l) MCANet (96.0%). (m) AMSNet (99.1%). Blue-green area represents background and blue area represents buildings.

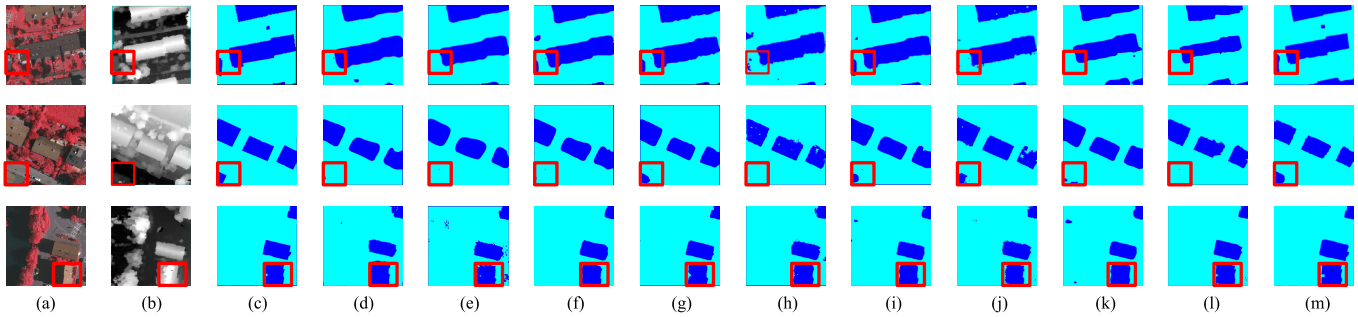


Fig. 7. Qualitative comparison of building extraction for different methods on the Vaihingen dataset. (a) Optical. (b) LiDAR. (c) Ground truth. (d) ResUNet++ (94.1%). (e) PSPNet (94.0%). (f) Deeplab V3+ (96.2%). (g) AMSNet-single (96.8%). (h) FuseNet (96.1%). (i) RedNet (96.8%). (j) HAFNet (96.4%). (k) $(MS)^2$-Net (95.3%). (l) MCANet (96.5%). (m) AMSNet (98.5%). Blue-green area represents background and blue area represents buildings.

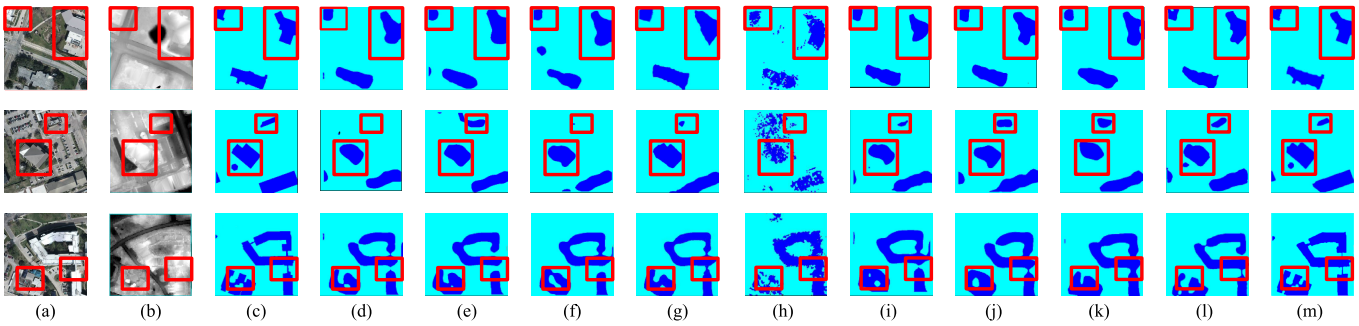


Fig. 8. Qualitative comparison of building extraction for different methods on the DFC 2018 dataset. (a) Optical. (b) LiDAR. (c) Ground truth. (d) ResUNet++ (91.6%). (e)PSPNet (88.3%). (f) Deeplab V3+ (86.0%). (g) AMSNet-single (93.5%). (h) FuseNet (83.2%). (i) RedNet (95.6%). (j) HAFNet (93.7%). (k) $(MS)^2$-Net (86.4%). (l) MCANet (96.8%). (m) AMSNet (98.9%). Blue-green area represents background and blue area represents buildings.

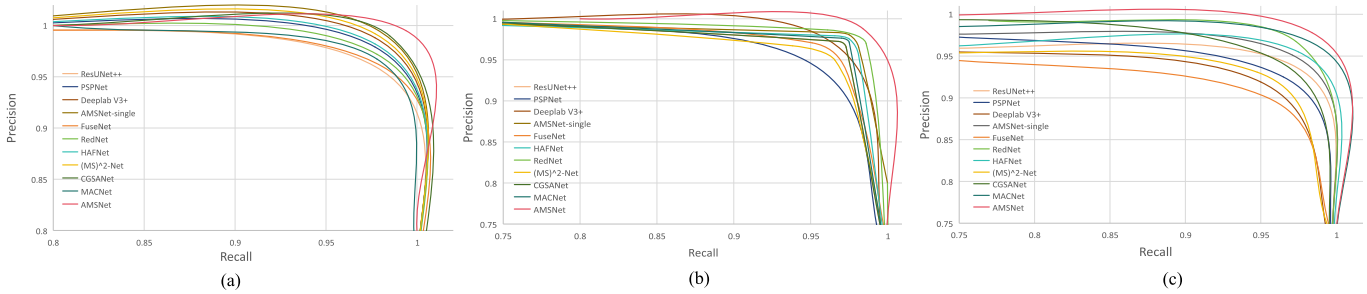


Fig. 9. P–R curves of ResUNet++, PSPNet, Deeplab V3+, AMSNet-single, FuseNet, RedNet, HAFNet, $(MS)^2$-Net, MCANet, and the proposed ASMNet on (a) Potsdam, (b) Vaihingen, and (c) DFC2018 datasets.

and MIoU compared with the other methods. For the Potsdam dataset, the OA, FWIoU, and MIoU of AMSNet are, respectively, 1.4%, 1.8%, and 1.8% higher than the best dual-branch competitor $(MS)^2$-Net. For the Vaihingen dataset, the OA, FWIoU, and MIoU of AMSNet outperform the best dual-branch competitor RedNet by 1.7%, 2.8%, and 20.8%,

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TABLE II
EVALUATION OF METRIC SCORES (%) OF SEMANTIC SEGMENTATION FOR THREE DATASETS

Dataset	Metrics	Single-branch networks				Dual-branch networks					
		ResUNet++ [36]	PSPNet [37]	Deeplab V3+ [38]	AMSNet-single	FuseNet [39]	RedNet [40]	HAFNet [16]	$(MS)^2$-Net [41]	MCANet [42]	AMSNet
Potsdam	OA	77.2±0.927	82.0±0.522	84.5±1.425	87.1±1.131	80.7±0.502	87.1±1.095	84.5±1.254	87.6±0.563	89.1±0.414	**90.5±0.368**
	FWIOU	64.4±1.315	70.6±1.006	74.9±2.827	77.2±2.207	70.0±0.983	78.3±1.844	74.7±2.104	78.9±0.957	81.1±0.863	**82.9±0.932**
	MIOU	54.7±2.330	58.7±1.553	59.6±3.960	62.2±2.421	57.0±1.520	65.7±3.373	59.3±3.735	69.0±1.648	70.3±1.203	**72.1±1.478**
Vaihingen	OA	88.3±0.294	89.8±1.029	85.2±1.057	91.7±0.157	76.9±0.663	92.2±0.584	78.9±0.212	90.6±0.753	92.4±0.388	**94.1±0.312**
	FWIOU	80.0±0.549	82.0±2.660	75.4±2.029	85.4±0.543	63.9±1.175	86.1±1.003	66.3±0.377	83.4±1.133	86.9±0.555	**89.7±0.690**
	MIOU	62.3±0.848	70.9±3.308	56.8±1.861	79.8±0.876	49.1±1.722	73.4±1.665	51.4±0.424	72.6±1.320	79.3±0.774	**82.1±0.705**
DFC2018	OA	79.8±0.741	77.5±1.315	76.7±1.900	81.3±0.224	69.6±1.583	81.6±1.320	72.3±1.397	80.5±0.516	84.5±1.532	**86.3±0.472**
	FWIOU	65.7±1.442	63.2±2.546	62.5±3.509	69.6±0.631	54.2±2.615	70.3±1.322	56.5±2.356	68.5±0.731	74.5±1.564	**77.6±1.355**
	MIOU	49.2±3.412	47.1±3.148	46.0±4.178	52.2±2.131	38.0±3.204	57.6±2.009	40.1±3.068	56.3±1.320	60.5±2.266	**65.1±2.243**

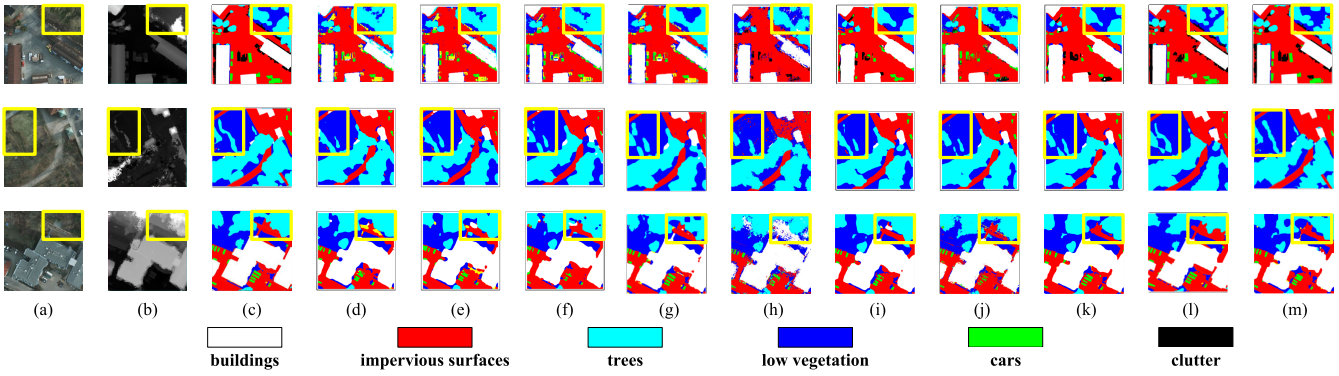


Fig. 10. Qualitative comparison of semantic segmentation for different methods on the Potsdam dataset. (a) Optical. (b) LiDAR. (c) Ground truth. (d) ResUNet++ (77.2%). (e) PSPNet (82.0%). (f) Deeplab V3+ (84.5%). (g) AMSNet-single (87.1%). (h) FuseNet (80.7%). (i) RedNet (97.1%). (j) HAFNet (84.5%). (k) $(MS)^2$-Net (87.6%). (l) MCANet (89.1%). (m) AMSNet (90.5%).

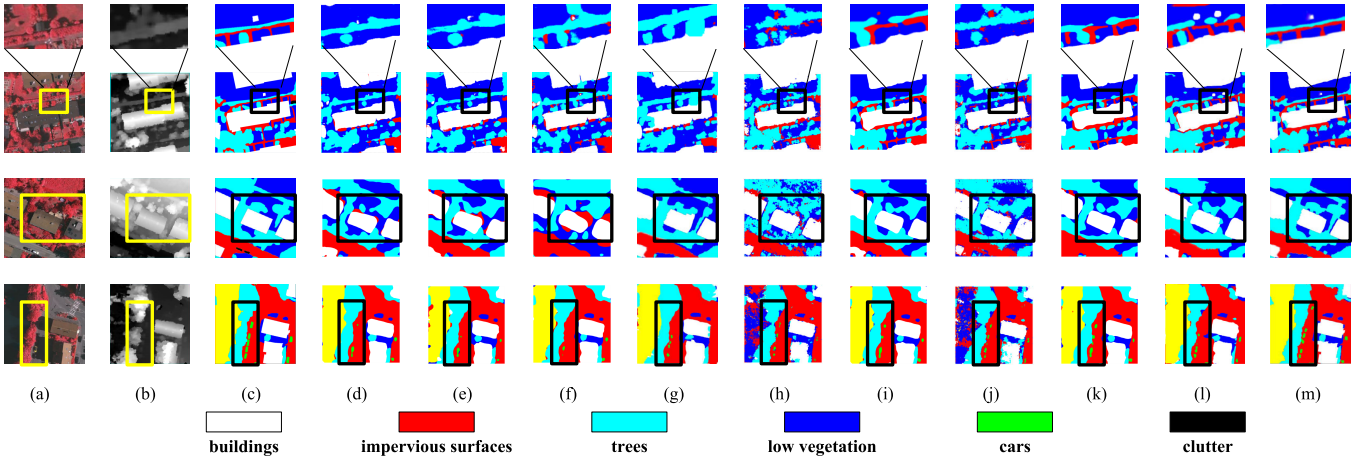


Fig. 11. Qualitative comparison of semantic segmentation for different methods on the Vaihingen dataset. (a) Optical. (b) LiDAR. (c) Ground truth. (d) ResUNet++ (88.3%). (e) PSPNet (89.8%). (f) Deeplab V3+ (85.2%). (g) AMSNet-single (91.7%). (h) FuseNet (76.9%). (i) RedNet (92.2%). (j) HAFNet (78.9%). (k) $(MS)^2$-Net (90.6%). (l) MCANet (92.4%). (m) AMSNet (94.1%).

respectively. For the DFC2018 dataset, the OA, FWIoU, and MIoU of AMSNet are significantly higher than those of RedNet by 1.8%, 3.1%, and 4.6%, respectively. Furthermore, the metrics of AMSNet-single also have the highest scores among single-branch methods for semantic segmentation. At the same time, AMSNet achieves significant improvements in semantic segmentation performance compared with AMSNet-single across all the three datasets, demonstrating the critical role of LiDAR data in enhancing overall segmentation accuracy. For the Potsdam dataset, AMSNet improves OA, FWIoU, and MIoU by 3.4%, 5.7%, and 9.9%, respectively, over AMSNet-single. For the Vaihingen dataset, OA, FWIoU, and MIoU increase by 2.4%, 4.3%, and 2.3%, respectively. For the more complex DFC2018 dataset, the improvements in OA, FWIoU, and MIoU are even more pronounced, reaching 5%, 8%, and 12.9%, respectively. These results demonstrate the superior performance of AMSNet in semantic segmentation on various datasets.

To exemplify the generalization of the proposed AMSNet, Figs. 10–12 show the results of the semantic segmentation of

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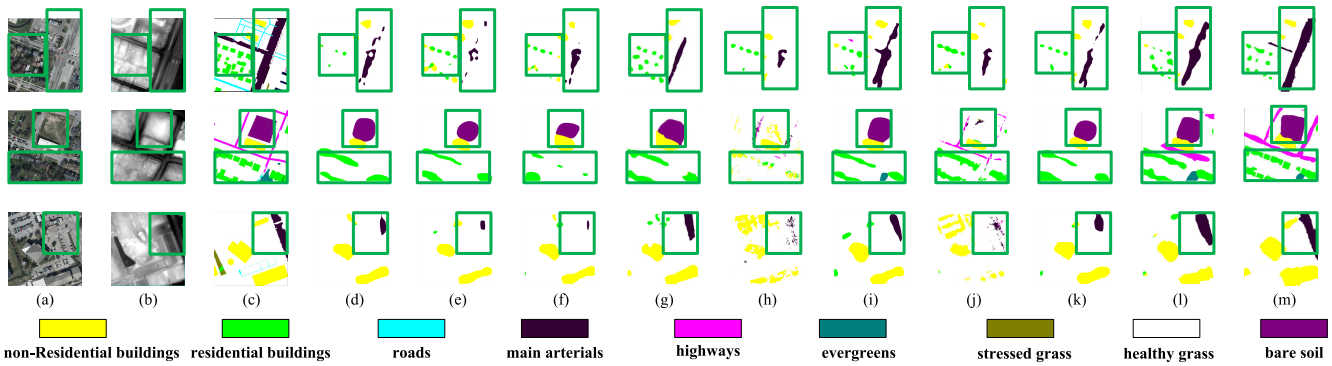


Fig. 12. Qualitative comparison of semantic segmentation for different methods on the DFC2018 dataset. (a) Optical. (b) LiDAR. (c) Ground truth. (d) ResUNet++ (79.8%). (e)PSPNet (77.5%). (f) Deeplab V3+ (96.7%). (g) AMSNet-single (81.3%). (h) FuseNet (69.6%). (i) RedNet (81.6%). (j) HAFNet (72.3%). (k) $(MS)^2$-Net (80.5%). (l) MCANet (84.5%). (m) AMSNet (86.3%).

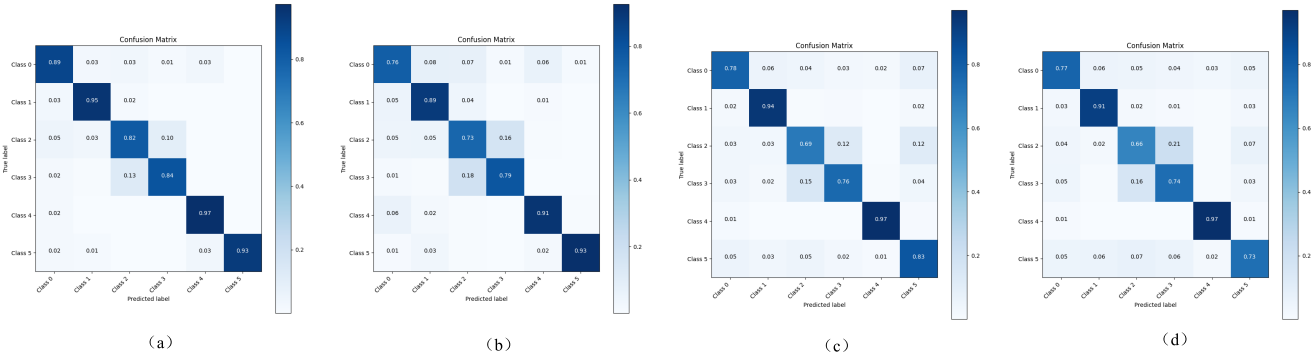


Fig. 13. Comparison of confusion matrices for single-modality and multimodality semantic segmentation on the Vaihingen and Potsdam datasets. (a) Vaihingen—multimodality. (b) Vaihingen—single-modality. (c) Potsdam—multimodality. (d) Potsdam—single-modality. Class 0: *Impervious surfaces*. Class 1: *Buildings*. Class 2: *Low vegetation*. Class 3: *Trees*. Class 4: *Car*. Class 5: *Clutter*.

all the networks. It can be seen that the proposed ASMNet provides the most accurate and smooth maps for all three datasets compared with other methods. Taking Fig. 11 Vaihingen dataset as an example, good segmentation results are achieved for large areas, especially with respect to buildings and trees. However, as far as local details are concerned, the robustness and effectiveness of the comparative models are not sufficient to accurately identify clutter, cars, and low vegetation. For example, as shown in the box in the first row of segmentation maps, buildings spread over large areas of low vegetation can be accurately segmented by ASMNet, but the segmentation performance of all other comparison networks is insufficient. This is also reflected in the following two cases. As in the second row of maps, segmenting low vegetation and number and the third row of maps segmenting out cars in a large area of impervious surface, the proposed ASMNet outperforms the other networks. Fig. 13 shows the single-modality and multimodality semantic segmentation results on the Vaihingen and Potsdam datasets. For the Vaihingen dataset, the accuracy of *buildings* improved by 6%, and *trees* improved by 5%. For the Potsdam dataset, the accuracy of *buildings* improved by 3%, and that of *trees* improved by 2%, demonstrating the advantage of multimodality in enhancing the classification performance of *buildings* and *trees*.

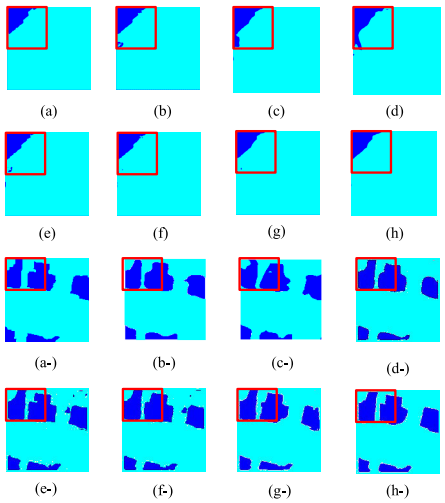


Fig. 14. Potsdam dataset building extraction maps for ablation experiments are plotted. (a) Ground truth. (b) Network A (97.6%). (c) Network B (97.3%). (d) Network C (97.4%). (e) Network D (97.8%). (f) Network E (97.7%). (g) Network F (98.1%). (h) AMSNet (99.1%). (a-) Ground truth. (b-) Network A- (94.6%). (c-) Network B- (93.5%). (d-) Network C- (93.9%). (e-) Network D- (96.7%). (f-) Network E- (96.4%). (g-) Network F- (98.0%). (h-) AMSNet- (98.8%). Blue-green area represents background and blue area represents buildings.

B. Ablation Experiments

To measure the role of the three main components (i.e., the S^2AF module, the RFL module, and the SGF module) in AMSNet, seven different networks were designed. Networks A, B, and C use only one component to build the model. Networks D, E, and F use two components to build the model, which was compared with the proposed AMSNet to determine

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TABLE III

EVALUATION OF METRIC SCORES (%) FOR BUILDING EXTRACTION ABLATION EXPERIMENTS

		S^2AF	RFL	SGF	OA	FWIoU	MIoU
Potsdam	Network A	✓			97.6	95.5	94.9
	Network B		✓		97.3	94.8	94.1
	Network C			✓	97.4	95.0	94.3
	Network D	✓	✓		97.8	95.8	95.2
	Network E		✓	✓	97.7	95.7	95.0
	Network F	✓		✓	98.1	96.4	95.9
	AMSNet	✓	✓	✓	**99.1**	**97.9**	**97.4**
	Network A-	✓			94.6	90.2	84.5
	Network B-		✓		93.5	88.4	82.9
	Network C-	✓	✓		93.9	89.9	86.2
	Network D-		✓	✓	96.7	93.8	89.6
	Network E-	✓		✓	96.4	93.4	89.2
	Network F-	✓		✓	98.0	96.2	93.3
	AMSNet-	✓	✓	✓	**98.8**	**97.7**	**94.6**
Vaihingen	Network A	✓			97.3	95.0	94.0
	Network B		✓		95.7	91.9	90.1
	Network C			✓	97.1	94.6	93.4
	Network D	✓	✓		97.5	95.5	94.7
	Network E		✓	✓	97.4	95.1	94.0
	Network F	✓		✓	97.9	96.0	95.3
	AMSNet	✓	✓	✓	**98.5**	**97.0**	**96.4**
	Network A-	✓			94.3	89.5	85.7
	Network B-		✓		92.8	86.7	81.8
	Network C-			✓	93.9	88.7	85.1
	Network D-		✓	✓	95.3	91.2	87.9
	Network E-		✓	✓	94.4	89.7	86.3
	Network F-	✓		✓	95.6	91.9	88.1
	AMSNet-	✓	✓	✓	**96.2**	**92.7**	**90.3**
DFC 2018	Network A	✓			97.4	95.1	91.2
	Network B		✓		95.5	91.8	85.9
	Network C			✓	97.1	94.4	90.5
	Network D	✓	✓		97.7	96.3	93.4
	Network E		✓	✓	97.4	95.6	91.9
	Network F	✓		✓	98.3	96.8	93.6
	AMSNet	✓	✓	✓	**98.9**	**97.7**	**94.2**
	Network A-	✓			96.1	92.8	86.9
	Network B-		✓		94.5	90.1	84.1
	Network C-			✓	95.8	92.2	87.1
	Network D-	✓	✓		96.6	93.7	89.2
	Network E-		✓	✓	96.3	93.2	88.3
	Network F-	✓		✓	97.1	94.4	90.6
	AMSNet-	✓	✓	✓	**97.8**	**95.8**	**93.0**

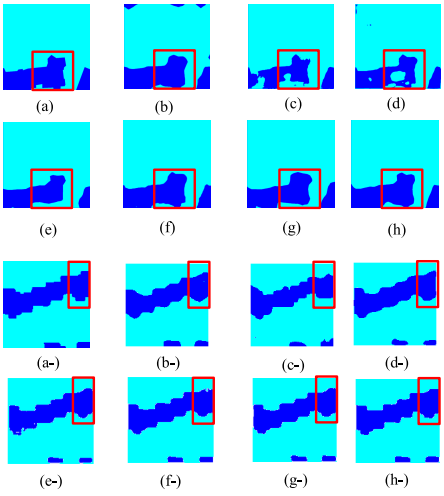


Fig. 15. Vaihingen dataset building extraction maps for ablation experiments are plotted. (a) Ground truth. (b) Network A (97.3%). (c) Network B (95.7%). (d) Network C (97.1%). (e) Network D (97.5%). (f) Network E (97.4%). (g) Network F (97.9%). (h) AMSNet (98.5%). (a-) Ground truth. (b-) Network A- (94.3%). (c-) Network B- (92.8%). (d-) Network C- (93.9%). (e-) Network D- (95.3%). (f-) Network E- (94.4%). (g-) Network F- (95.6%). (h-) AMSNet- (96.2%). Blue-green area represents background and blue area represents buildings.

TABLE IV

EVALUATION OF METRIC SCORES (%) FOR SEMANTIC SEGMENTATION ABLATION EXPERIMENTS

		S^2AF	RFL	SGF	OA	FWIoU	MIoU
Potsdam	Network A	✓			87.9	78.7	62.9
	Network B		✓		88.9	80.0	63.2
	Network C			✓	89.0	80.3	69.0
	Network D	✓	✓		89.1	80.5	69.7
	Network E		✓	✓	89.5	81.3	71.8
	Network F	✓		✓	89.3	80.9	71.4
	AMSNet	✓	✓	✓	**90.5**	**82.9**	**72.1**
	Network A-	✓			87.3	78.5	62.2
	Network B-		✓		88.7	79.7	63.0
	Network C-			✓	88.9	80.1	68.7
	Network D-	✓	✓		89.1	80.4	69.3
	Network E-		✓	✓	89.7	80.6	70.9
	Network F-	✓		✓	89.0	80.2	68.9
	AMSNet-	✓	✓	✓	**90.2**	**82.1**	**71.9**
Vaihingen	Network A	✓			83.1	70.4	61.7
	Network B		✓		84.0	72.0	64.0
	Network C			✓	89.8	81.6	76.5
	Network D	✓	✓		88.4	79.3	73.4
	Network E		✓	✓	92.4	86.9	80.3
	Network F	✓		✓	90.3	83.5	77.3
	AMSNet	✓	✓	✓	**94.1**	**89.7**	**82.1**
	Network A-	✓			82.7	69.3	60.7
	Network B-		✓		83.8	71.7	63.7
	Network C-			✓	89.5	81.2	76.0
	Network D-	✓	✓		89.0	79.2	73.3
	Network E-		✓	✓	91.1	85.4	79.6
	Network F-	✓		✓	90.7	83.8	77.9
	AMSNet-	✓	✓	✓	**93.7**	**87.8**	**80.9**
DFC 2018	Network A	✓			79.6	65.0	47.4
	Network B		✓		80.5	66.6	50.3
	Network C			✓	82.5	69.7	51.1
	Network D	✓	✓		84.2	71.8	53.7
	Network E		✓	✓	85.6	74.5	59.8
	Network F	✓		✓	84.7	72.7	56.1
	Network G	✓		✓	85.1	73.4	59.9
	AMSNet	✓	✓	✓	**86.3**	**77.6**	**65.1**
	Network A-	✓			78.6	63.2	45.5
	Network B-		✓		79.4	65.9	47.2
	Network C-			✓	80.3	67.7	49.9
	Network D-	✓	✓		82.3	69.9	52.6
	Network E-		✓	✓	84.2	71.8	57.7
	Network F-	✓		✓	83.7	70.6	54.9
	AMSNet-	✓	✓	✓	**85.1**	**73.4**	**59.9**

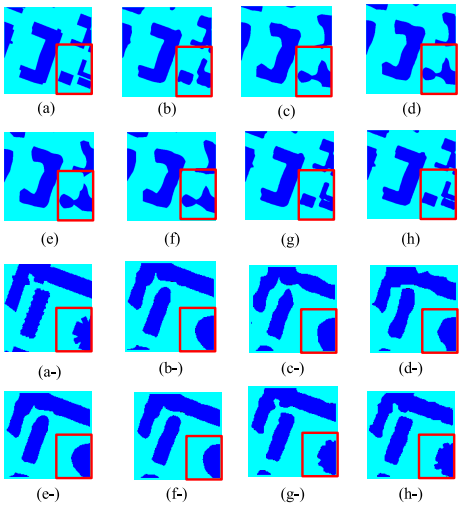


Fig. 16. DFC2018 dataset building extraction maps for ablation experiments are plotted. (a) Ground truth. (b) Network A (97.4%). (c) Network B (95.5%). (d) Network C (97.1%). (e) Network D (97.7%). (f) Network E (97.4%). (g) Network F (98.3%). (h) AMSNet (98.9%). (a-) Ground truth. (b-) Network A- (96.1%). (c-) Network B- (94.5%). (d-) Network C- (95.8%). (e-) Network D (96.6%). (f-) Network E- (96.3%). (g-) Network F (97.1%). (h-) AMSNet- (97.8%). Blue-green area represents background and blue area represents buildings.

the effect of the different components. In addition, to further examine the role of the three main components with different backbones, we replaced the ResNet-34 backbone in AMSNet with the backbone from $(MS)^2$-Net. The corresponding

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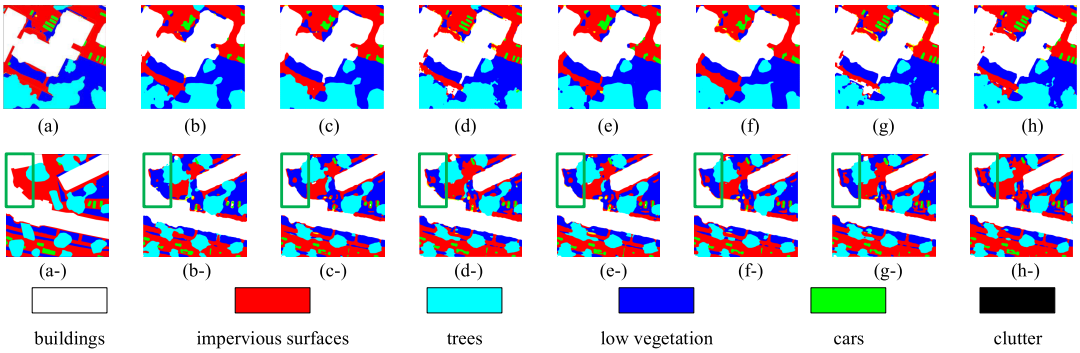


Fig. 17. Potsdam dataset semantic segmentation maps for ablation experiments are plotted. (a) Ground truth. (b) Network A (87.9%). (c) Network B (88.9%). (d) Network C (89.0%). (e) Network D (89.1%). (f) Network E (89.5%). (g) Network F (89.3%). (h) AMSNet (90.5%). (a-) Ground truth. (b-) Network A- (87.3%). (c-) Network B- (88.7%). (d-) Network C- (88.9%). (e-) Network D- (89.1%). (f-) Network E- (89.7%). (g-) Network F- (89.0%). (h-) AMSNet- (90.2%).

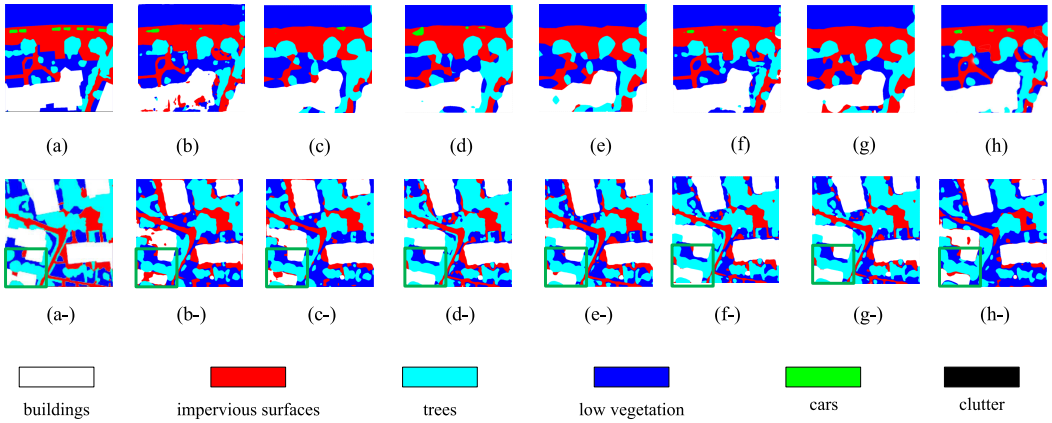


Fig. 18. Vaihingen dataset semantic segmentation maps for ablation experiments are plotted. (a) Ground truth. (b) Network A (83.1%). (c) Network B (84.0%). (d) Network C (89.8%). (e) Network D (88.4%). (f) Network E (92.4%). (g) Network F (90.3%). (h) AMSNet (94.1%). (a-) Ground truth. (b-) Network A- (82.7%). (c-) Network B- (83.8%). (d-) Network C- (89.5%). (e-) Network D- (89.0%). (f-) Network E- (91.1%). (g-) Network F- (90.7%). (h-) AMSNet- (93.7%).

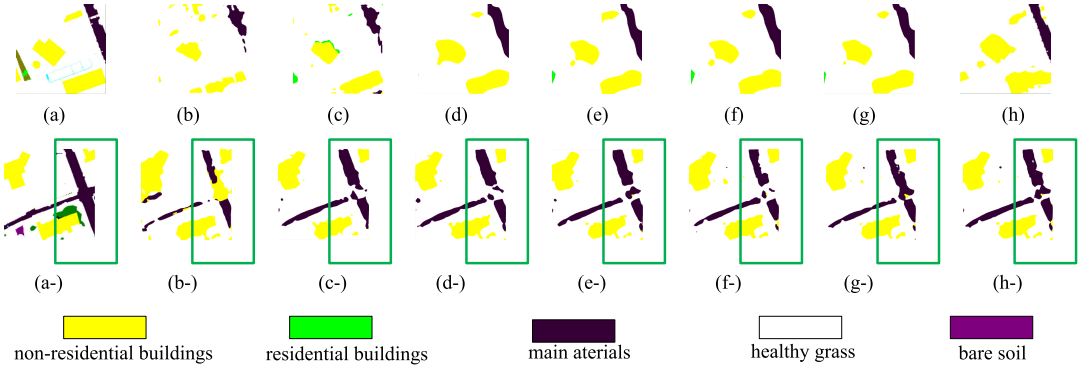


Fig. 19. DFC2018 dataset semantic segmentation maps for ablation experiments are plotted. (a) Ground truth. (b) Network A (79.6%). (c) Network B (80.5%). (d) Network C (82.5%). (e) Network D (84.2%). (f) Network E (85.6%). (g) Network F (84.7%). (h) AMSNet (86.3%). (a-) Ground truth. (b-) Network A- (78.6%). (c-) Network B- (79.4%). (d-) Network C- (80.3%). (e-) Network D- (82.3%). (f-) Network E- (84.2%). (g-) Network F- (83.7%). (h-) AMSNet- (85.1%).

networks are denoted as Networks A-, B-, C-, D-, E-, F-, and AMSNet-.

As shown in Table III, the results of the seven networks for the building extraction task show a general consistency in the impact of the components across three datasets. In terms of the single-module networks, Network A that performs better than Network B and Network C, which gives a rough indication that the S^2AF modules contribute more to the performance of building extraction. In the case of two-module networks, the impact of each component is further analyzed using the Potsdam dataset as an example. The two-module networks further indicate that S^2AF module contributes more to the performance of building extraction. When we used the conventional 2D convolution instead of the SGF module In Network D, the OA/FWIoU/MIoU of AMSNet is 1.3%/2.1%/2.2% higher than that of Network D. In Network E, S^2AF is

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TABLE V
COMPUTATION OF TRAINING TIME, TESTING TIME, FLOPS, PARAMETERS, AND MEMORY CONSUMPTION FOR THREE DATASETS

Dataset	Metrics	Single-branch networks				Dual-branch networks					
		ResUNet++ [36]	PSPNet [37]	Deeplab V3+ [38]	AMSNet -single	FuseNet [39]	RedNet [40]	HAFNet [16]	$(MS)^2$-Net [41]	MCANet [42]	AMSNet
-	FLOPs (G)	98.87	249.21	139.31	284.29	376.54	132.27	737.19	136.78	432.49	409.99
-	Params (M)	4.06	27.50	59.54	108.93	44.17	81.94	88.98	13.73	53.35	153.30
-	Memory (M)	2740	4018	5788	2052	4136	3800	4784	3102	4100	4222
Potsdam	Training Time (s)	90.76	105.09	97.52	109.72	140.89	221.12	214.44	196.89	230.51	229.41
	Testing Time (s)	33.20	40.29	37.78	43.82	59.30	36.96	77.93	36.63	71.77	69.89
Vaihingen	Training Time (s)	28.71	41.56	31.13	49.32	44.24	70.08	68.52	65.89	75.83	73.95
	Testing Time (s)	3.62	4.24	3.78	4.93	5.52	3.67	6.82	3.71	6.05	6.01
DFC2018	Training Time (s)	2.14	2.78	2.64	2.91	2.97	5.07	3.98	4.23	5.32	4.93
	Testing Time (s)	0.19	0.26	0.22	0.32	0.41	0.21	0.80	0.22	0.81	0.60

replaced by the traditional summation and data fusion strategy. AMSNet's OA/FWIoU/MIoU improves by 1.4%/2.1%/2.2%, compared with Network E. To evaluate the impact of the RFL module on the performance of AMSNet, we compare the experimental results of Network F and AMSNet. It can be seen that the OA/FWIoU/MIoU of AMSNet is improved by 1.0%/1.5%/1.5% compared with Network F. Although the performance of building extraction solely using RFL or SGF modules is not ideal, consideration of the characteristics of these three modules and combining them in the proposed AMSNet will yield satisfactory results. Furthermore, when the ResNet-34 backbone in AMSNet was replaced with the backbone from $(MS)^2$-Net, these networks exhibited similar performance, validating the effect of the three components.

Accuracy metrics are the most effective means of quantitatively evaluating results. In the ablation experiments discussed above, we present numerical accuracies and offer concise descriptions of the results. To enhance readability, we provide visual representation and description of the role and impact of each module, as depicted in Figs. 14–16. When compared with Networks A–F (Networks A- to F-), the proposed AMSNet (AMSNet-) that incorporates three components exhibits superior performance in building extraction maps.

In addition, to assess the generalization of AMSNet, we conducted a semantic segmentation task for ablation experiments. The results varied slightly from the building extraction task. As shown in Table IV, we designed seven networks identical to those in Table III to measure the impact of the three main components in AMSNet and AMSNet-. As can be seen from the table, Network C possesses optimal perform of semantic segmentation compared with Networks A and B, which indicates SGF modules contribute more than S^2AF and RFL modules. The similar situation lies in the two-module networks. That is, Networks E and F used SGF modules which outperformed Network D for semantic segmentation. To further evaluate the impact of the SGF module on the performance of AMSNet, we compare Network D with AMSNet, whose OA/FWIoU/MIoU is higher than that of Network D by 5.7%/10.4%/8.7%. In addition, when the ResNet-34 backbone in AMSNet was substituted with the backbone from $(MS)^2$-Net, the networks demonstrated comparable performance, confirming the impact of the three components. Furthermore, similar to the ablation of semantic segmentation, we have presented the segmentation maps. As shown in Figs. 17–19, it can be seen that the proposed AMSNet (AMSNet-) used three components works best compared with other architectures.

C. Number of Parameters and Calculation Time

To illustrate the effectiveness of various methods, we present training time, testing time, floating-point operations (FLOPs), the number of parameters (Params), and the memory consumption (M) in Table V. Typically, single-branch methods exhibit lower computational complexity than dual-branch methods, as their model design does not incorporate LiDAR feature learning. In this study, AMSNet is configured as a dual-branch method that achieves outstanding building extraction performance. Despite being designed for superior performance, the computational complexity of AMSNet remains acceptable when compared with other state-of-the-art methods.

VI. CONCLUSION

The primary objective of this study is to perform building extraction and semantic segmentation using optical and LiDAR remote sensing images. We introduce a novel AMSNet using an encoding–decoding structure. The encoder component of AMSNet is constructed with optical and LiDAR streams, and multimodal feature fusion is achieved through the S^2AF module. Meanwhile, the decoder component incorporates upper level and lower level feature fusion, accomplished by merging higher level features of multimodal information with lower level features through the SGF module. In addition, network robustness is ensured by the RFL module. The proposed method is assessed using three publicly available optical remote sensing images along with the corresponding LiDAR datasets, covering urban structures with diverse sizes and poses. The experimental results demonstrate that AMSNet surpasses other fusion-based methods, and the effectiveness of all major components is confirmed through extensive ablation studies.

In future work, we will focus on the following directions: 1) acquiring additional datasets to explore segmentation performance in uncovered scenarios; and 2) improving the

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computational efficiency of our model, which currently surpasses existing models in accuracy, by developing lightweight semantic segmentation networks.

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